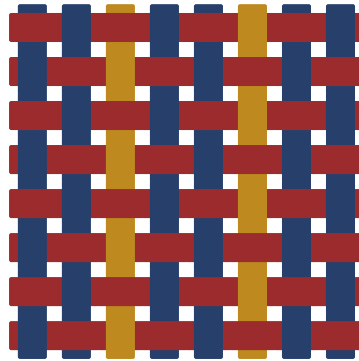




Entropy & Information

Fill-in study notes on Shannon's theory

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dyed in indigo, madder & weld ruled in iron-gall. woven on loom.

1 INFORMATION CONTENT

A random variable X with outcomes $x_1, \dots, x_n$ and probabilities $p_1, \dots, p_n$.

→ entropy

Definition 1.1 (Shannon, 1948). The **entropy** of X is defined as

$$H(X) = - \sum_{i=1}^n p_i \log_2 p_i$$

measured in **bits**.

Think of entropy as “average surprise.” If you already know the outcome, surprise is zero. If every outcome is equally likely, surprise is maximised.

The information content of a single outcome x_i is _____.

? Why $\log_2$ and not $\ln$?

Theorem 1.2 (Maximum entropy). Among all distributions on n outcomes, entropy is maximised by the _____ distribution, achieving $H = \log_2 n$ bits.

Proof. Apply Jensen’s inequality to the concave function $f(p) = -p \log p$.

[fill in: complete the Jensen argument]

1.1 Properties of Entropy

Key identities

Trigger. When comparing two sources, reach for relative entropy.

1. $H(X) \geq 0$ with equality iff X is deterministic ■■■■
2. $H(X) \leq \log_2 |\mathcal{X}|$ with equality iff X is uniform ■■■■
3. $H(X, Y) = H(X) + H(Y|X)$ (chain rule) ■■■■

Your turn. Compute $H(X)$ for a biased coin with $p = \frac{1}{4}$:

~~~~~  
What’s the chain rule for three variables?

## 1.2 The Identities (the information diagram)

| identity                                            | reading                                         |
|-----------------------------------------------------|-------------------------------------------------|
| $H(X, Y) = H(X) + H(Y X)$                           | <b>chain rule:</b> total = first + rest         |
| $I(X; Y) = H(X) + H(Y) - H(X, Y)$                   | shared = overlap of the two circles             |
| $I(X; Y) = D_{\text{KL}}(p_{XY} \parallel p_X p_Y)$ | information = distance from <b>independence</b> |
| $I(X; Y) \geq 0$ , with = 0 iff independent         | you never lose by looking at $Y$                |

## 2 MUTUAL INFORMATION

→ mi

**Definition 2.1.** The **mutual information** between  $X$  and  $Y$  is

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

Mutual information measures how much knowing  $Y$  tells you about  $X$ . It's symmetric: knowing  $X$  tells you the same amount about  $Y$ .

**Example 2.2.** Let  $X$  be a fair coin and  $Y = X$ . Then  $I(X; Y) = H(X) = 1$  bit. The copy gives you everything. [ entropy ↔ ]

**Lemma 2.3** (Non-negativity).  $I(X; Y) \geq 0$  with equality iff  $X$  and  $Y$  are independent.

*Remark* (Connection to KL divergence).  $I(X; Y) = D_{\text{KL}}(p_{X,Y} \parallel p_X p_Y)$ , which makes non-negativity immediate from Gibbs' inequality.

■ L03

**Corollary 2.4.** Data processing inequality: if  $X \rightarrow Y \rightarrow Z$  forms a Markov chain, then  $I(X; Z) \leq I(X; Y)$ .

*Proof.* Expand using chain rule for mutual information. The Markov condition  $p(z|x, y) = p(z|y)$  kills the extra term. ■